Lab Exam

Arpan Adhikari

Set A

Decoder 3*8:

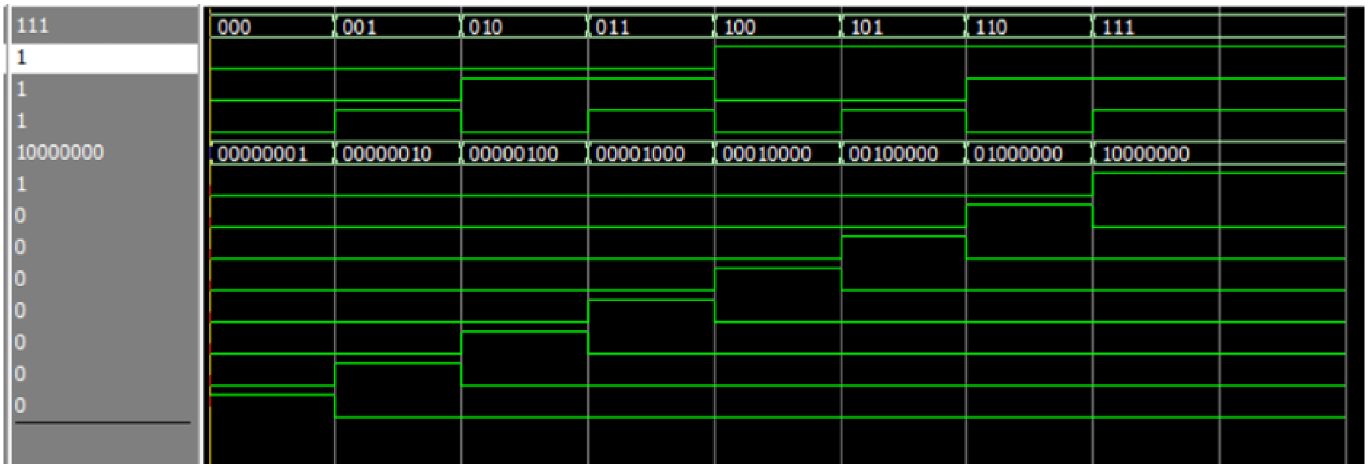
Behavioral:

```
library IEEE;
use IEEE.std_logic_1164.all;

entity arpan_decoder is
port(
   I : in std_logic_vector(2 downto 0);
   Y : out std_logic_vector(7 downto 0)
);
end arpan_decoder;

architecture behav_decoder of arpan_decoder is
begin
 process(I)
 begin
  case I is
   when "000" => Y <= "00000001";
   when "001" => Y <= "00000010";
   when "010" => Y <= "00000100";
   when "011" => Y <= "00001000";
   when "100" => Y <= "00010000";
   when "101" => Y <= "00100000";
   when "110" => Y <= "01000000";
   when "111" => Y <= "10000000";
   when others => Y <= "0";
end case;
 end process;
end behav_decoder;
```

Testbench:

```
library IEEE;
use IEEE.std_logic_1164.all;

entity arpan_test_decoder is
end arpan_test_decoder;

architecture test_decoder of arpan_test_decoder is
component arpan_decoder is
port(
   I : in std_logic_vector(2 downto 0);
   Y : out std_logic_vector(7 downto 0)
);
end component;

signal I : std_logic_vector(2 downto 0);
signal Y : std_logic_vector(7 downto 0);

begin
UUT: arpan_decoder port map
(
 I => I,
 Y => Y
);

stim_proc : process
begin
 I <= "000"; wait for 100 ps;
 I <= "001"; wait for 100 ps;
 I <= "010"; wait for 100 ps;
 I <= "011"; wait for 100 ps;
 I <= "100"; wait for 100 ps;
 I <= "101"; wait for 100 ps;
 I <= "110"; wait for 100 ps;
 I <= "111"; wait for 100 ps;
end process;
end test_decoder;
```

Waveform:


111
1
1
1
10000000
1
0
0
0
0
0
0
0
000
001
010
011
100
101
110
111
00000001
00000010
00000100
00001000
00010000
00100000
01000000
10000000

Mux 4*1:

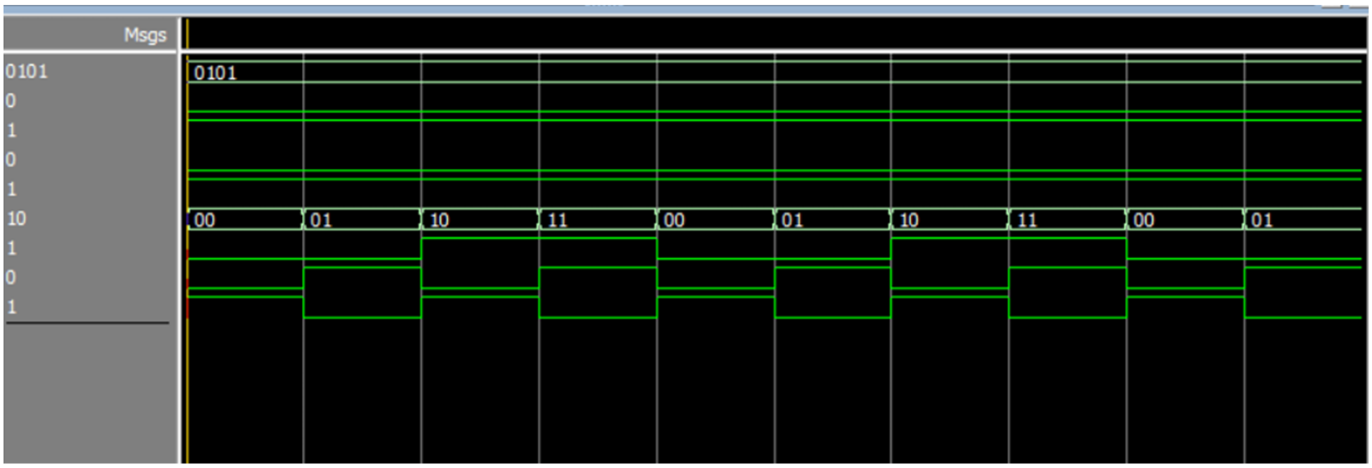
Behavioral:

```
library IEEE;
use IEEE.std_logic_1164.all;

entity arpan_mux is
port(
   I : in std_logic_vector(3 downto 0);
   S : in std_logic_vector(1 downto 0);
   Y : out std_logic
);
end arpan_mux;

architecture behav_mux of arpan_mux is
begin
 process(I)
 begin
  case S is
   when "00" => Y <= I(0);
   when "01" => Y <= I(1);
   when "10" => Y <= I(2);
   when "11" => Y <= I(3);
   when others => Y <= '0';
  end case;
end process;
end behav_mux;
```

Testbench:

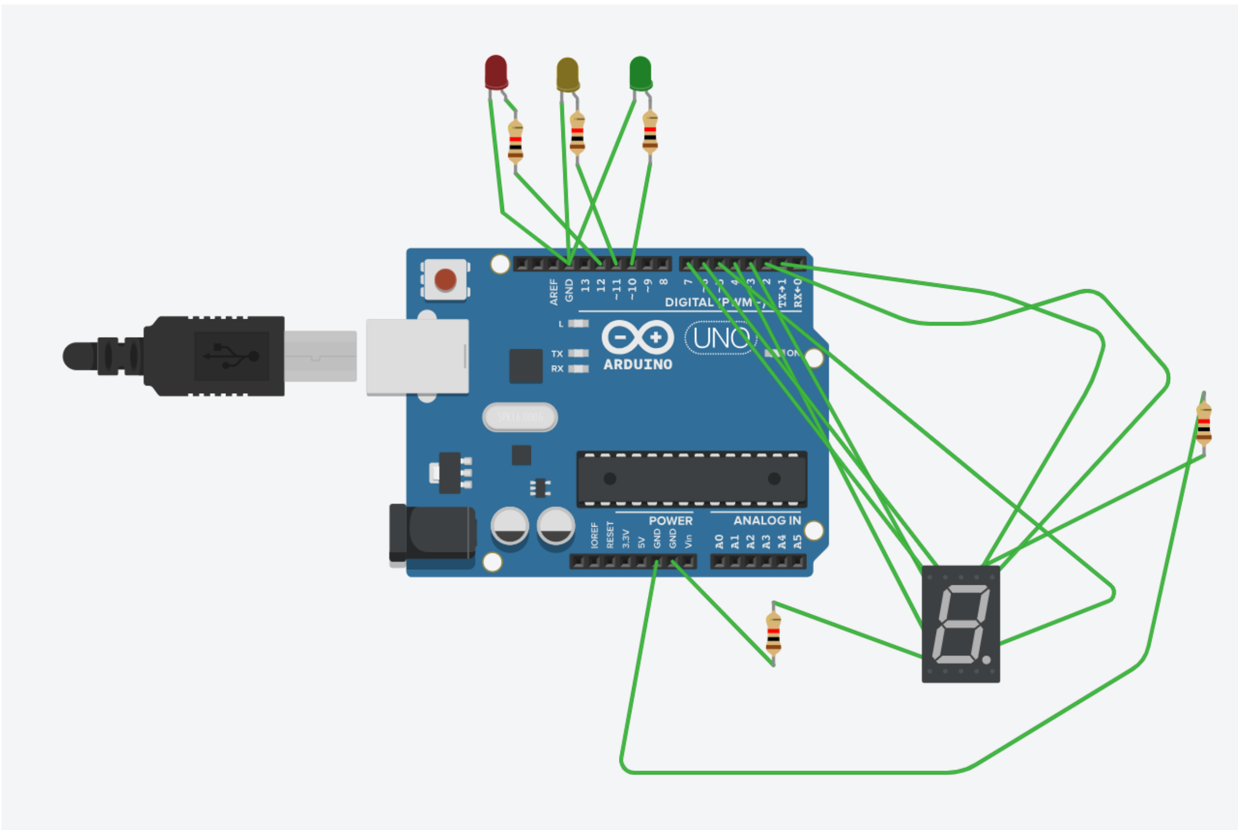
```vhdl
library IEEE;
use IEEE.std_logic_1164.all;

entity arpan_test_mux is
end arpan_test_mux;

architecture test_mux of arpan_test_mux is
component arpan_mux is
port(
  I : in std_logic_vector(3 downto 0);
  S : in std_logic_vector(1 downto 0);
  Y : out std_logic
);
end component;

signal I : std_logic_vector(3 downto 0) := "0101";
signal S : std_logic_vector(1 downto 0);
signal Y : std_logic;

begin
UUT: arpan_mux port map
(
 I => I,
 S => S,
 Y => Y
);

stim_proc : process
begin

 S <= "00"; wait for 100 ps;
 S <= "01"; wait for 100 ps;
 S <= "10"; wait for 100 ps;
 S <= "11"; wait for 100 ps;
end process;
end test_mux;
```

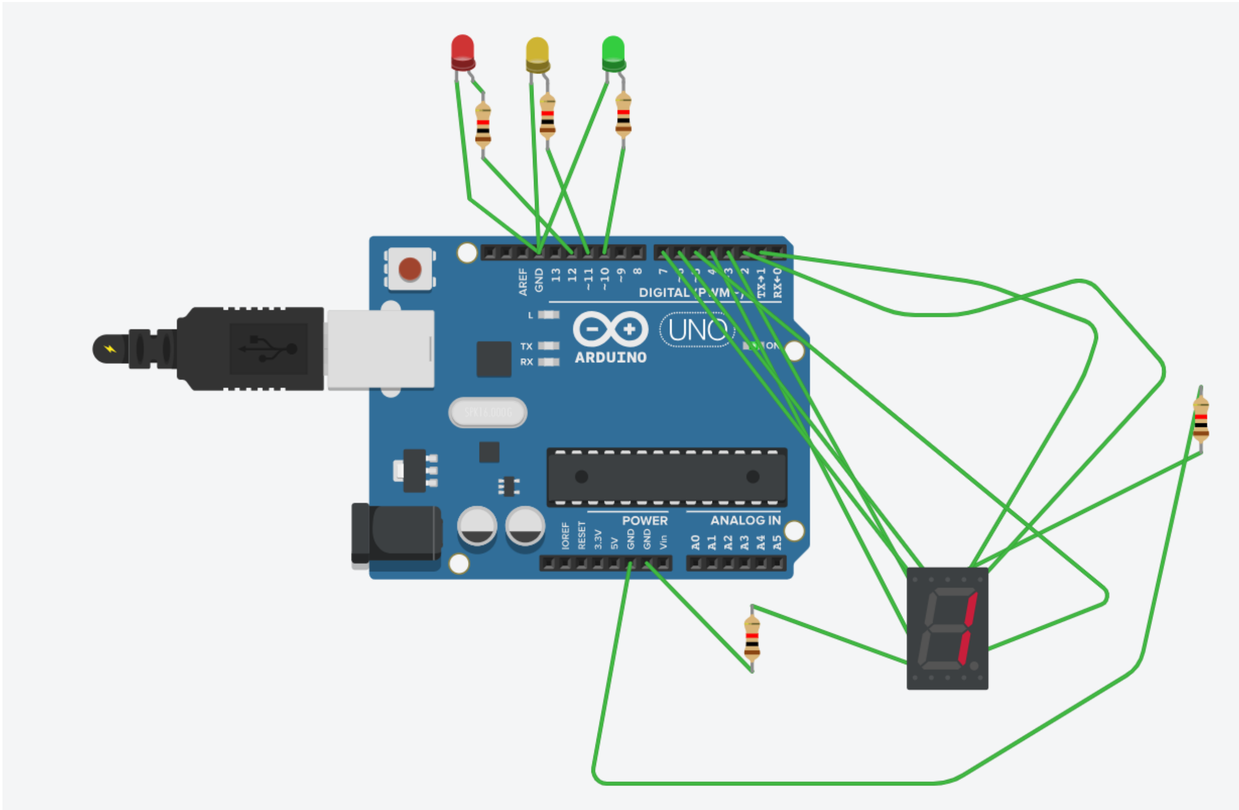
Waveform:



Tinkercad:

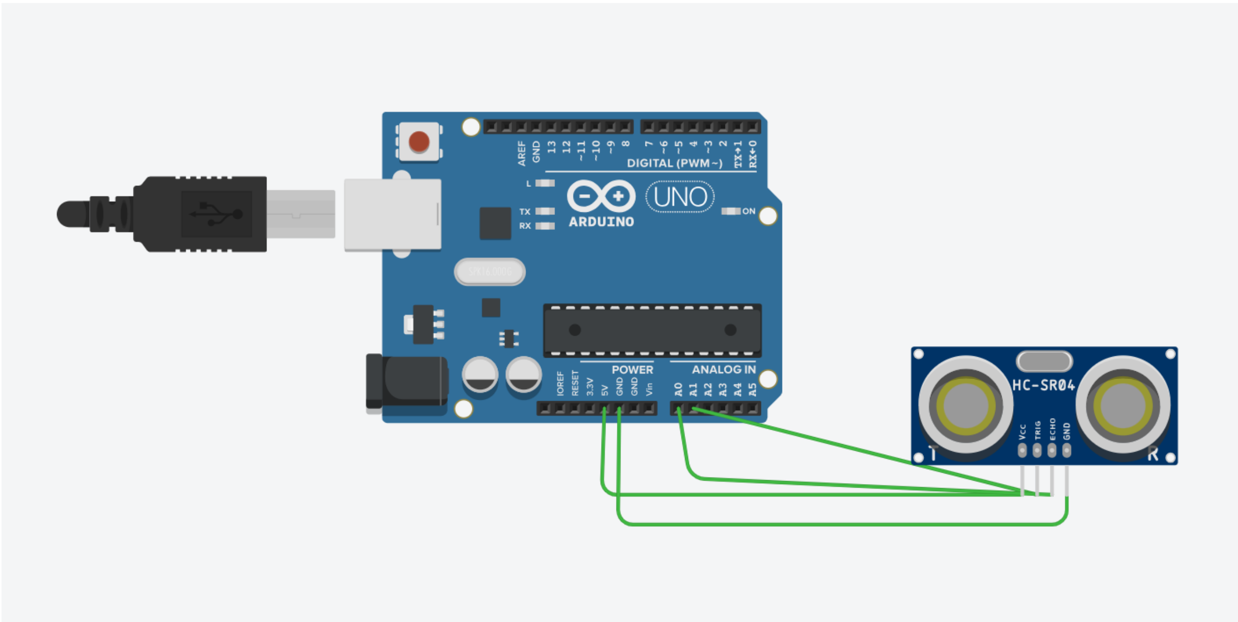
7 segment display with LED traffic light:

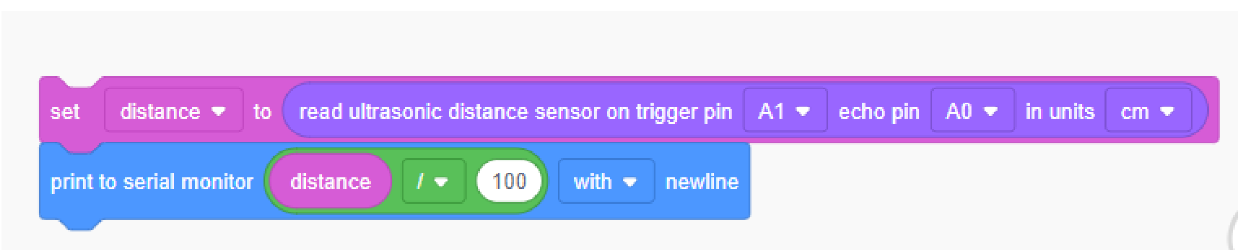


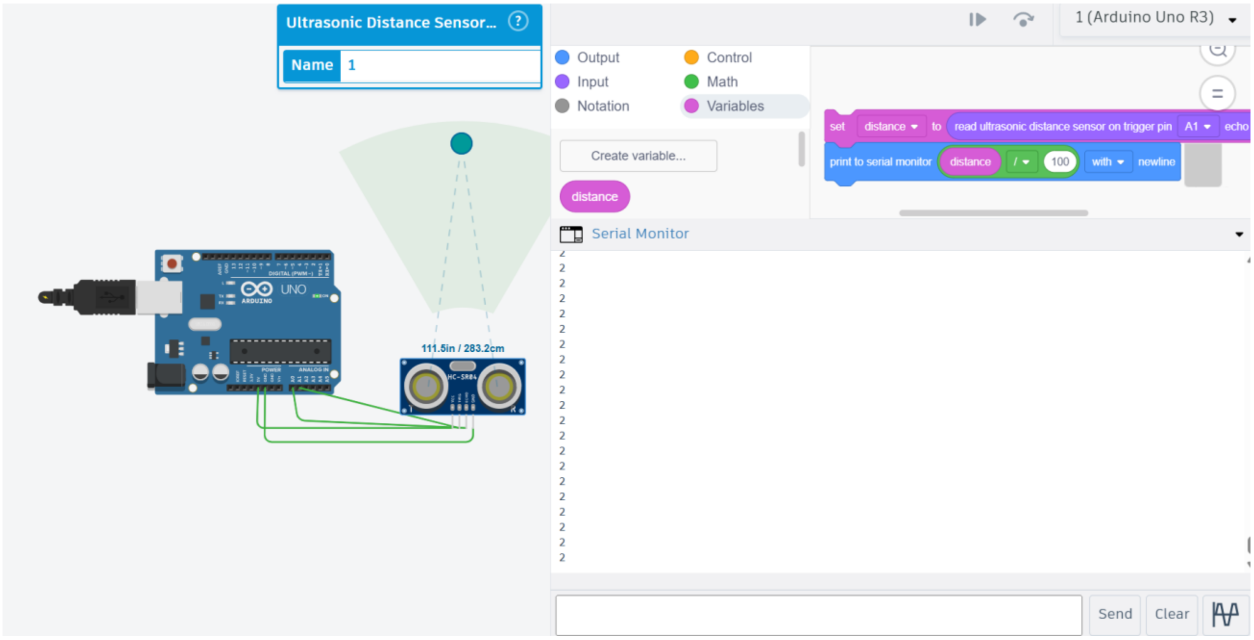


AREF
GND
13
12
~11
~10
~9
8
7
DIGITAL (PWM ~)
TX▸1
RX◂0
L
TX
RX
ARDUINO
UNO
ON
IOREF
RESET
3.3V
5V
GND
GND
Vin
POWER
ANALOG IN
A0
A1
A2
A3
A4
A5

Ultrasonic Sensor Interfacing to measure distance in meters:






Ultrasonic Distance Sensor...
Name 1
111.5in / 283.2cm
1 (Arduino Uno R3)
Output
Control
Input
Math
Notation
Variables
Create variable...
distance
set distance to read ultrasonic distance sensor on trigger pin A1 echo
print to serial monitor distance / 100 with newline
Serial Monitor
Send
Clear